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Integrated deep learning approach for generating cross-polarized images and analyzing skin melanin and hemoglobin distributions

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Abstract

Cross-polarized images are beneficial for skin pigment analysis due to the enhanced visualization of melanin and hemoglobin regions. However, the required imaging equipment can be bulky and optically complex. Additionally, preparing ground truths for training pigment analysis models is labor-intensive. This study aims to introduce an integrated approach for generating cross-polarized images and creating skin melanin and hemoglobin maps without the need for ground truth preparation for pigment distributions. We propose a two-component approach: a cross-polarized image generation module and a skin analysis module. Three generative adversarial networks (CycleGAN, pix2pix, and pix2pixHD) are compared for creating cross-polarized images. The regression analysis network for skin analysis is trained with theoretically reconstructed ground truths based on the optical properties of pigments. The methodology is evaluated using the VISIA VAESTRO clinical system. The cross-polarized image generation module achieved a peak signal-to-noise ratio of 35.514 dB. The skin analysis module demonstrated correlation coefficients of 0.942 for hemoglobin and 0.922 for melanin. The integrated approach yielded correlation coefficients of 0.923 for hemoglobin and 0.897 for melanin, respectively. The proposed approach

achieved a reasonable correlation with the professional system using actually captured images, offering a promising alternative to existing professional equipment without the need for additional optical instruments or extensive ground truth preparation.

Keywords: Skin pigments, Cross-polarized image, Generative adversarial networks, Deep learning

Introduction

Melanin and hemoglobin are the primary pigments in the skin. These pigments are related to skin tone and conditions such as melasma and freckles [1], as well as skin disorders like malignant melanoma and hemangiomas [2]. Accurate mapping of these pigments' distribution can aid in diagnosing and treating skin conditions and disorders [3], which makes it essential for professional diagnostic equipment to provide pigment maps [4, 5].

To discuss the imaging modalities and analysis methods for acquiring pigment maps, we first consider imaging modalities. For professional diagnostics requiring pigment information not visible to the naked eye, UV or cross-polarized images can be used. However, UV images capture fluorescence from porphyrins and sebum [6] and are strongly absorbed by both melanin and hemoglobin [7], making differentiation difficult. Cross-polarized images, on the other hand, minimize skin surface specular effects [8] and detect light from deeper skin layers, aiding in assessing melanin and hemoglobin distributions [5]. Despite their benefits, these modalities face practical challenges such as increased cost, equipment complexity, and maintenance needs. Some studies have used Conditional Generative Adversarial Networks (cGAN) to generate UV images [9], but we found no attempts to generate cross-polarized images for dermatological applications.

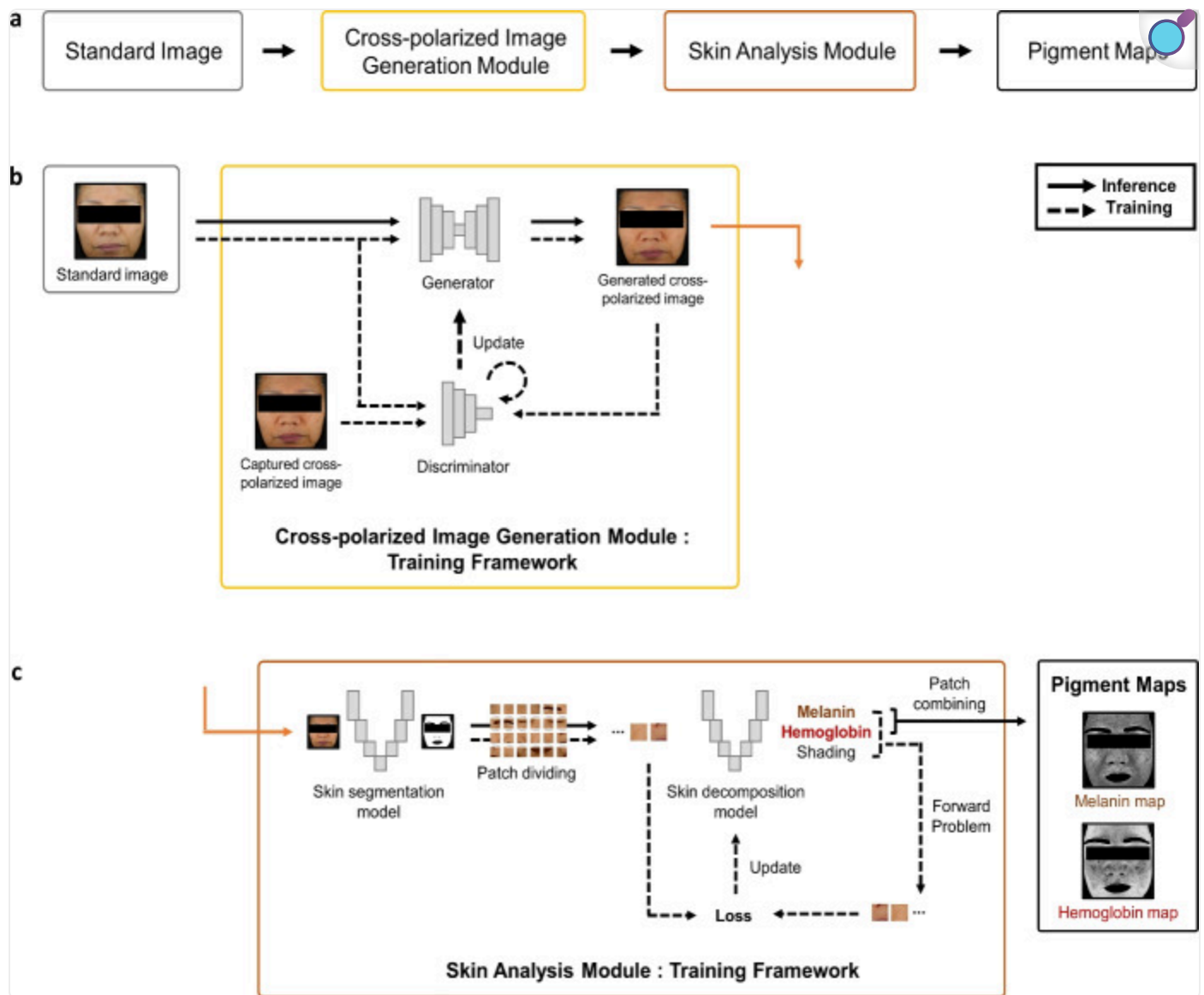
Next, most conventional approaches for acquiring pigment maps rely on image processing techniques, such as independent component analysis (ICA) on RGB images [10] or solving the inverse problem using spectral images [11]. While these methods provide reliable information for professional diagnostics [12], they require significant processing time for high-resolution images due to complex calculations for each pixel. Additionally, they are applicable only under specific acquisition conditions and need substantial pigment concentration variation within the image [13], presenting practical limitations. Although deep learning is gaining attention in dermatology, most studies focus on skin disease classification and segmentation [14, 15]. Training a regression network to generate pigment maps could reduce processing times, but creating ground truth for pigment maps is labor-intensive, time-consuming, and subjective.

In this study, we propose a deep learning-based approach for generating cross-polarized images and pigment maps, consisting of an image generation module and a skin analysis module. We compare three GAN models, trained with paired or unpaired datasets, to transform standard skin images into cross-polarized images. The skin analysis module then creates pigment maps from these images using a regression network trained with theoretically reconstructed ground truths based on the optical properties of pigments. Our methodology is evaluated against the VISIA system, widely used in clinics and research [4], to validate the performance of the individual modules and the integrated approach.

Materials and methods

The proposed approach, which consists of the cross-polarized image generation module and the skin analysis module, is illustrated in Fig. 1. While generative models require large amounts of training data and computing resources, there are practical limitations in collecting and processing high-quality images for dermatological applications. Therefore, we used GANs instead of diffusion models to effectively train under realistic conditions. For the skin analysis module, we used a lightweight version of the ResUNet++ network, which has demonstrated superior performance compared to standalone ResNet and UNet models in previous research [16].

Fig. 1.



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Schematic diagram of the integrated approach for generating cross-polarized images and creating melanin and hemoglobin maps. **a** The sequential modules generate the corresponding cross-polarized image from the standard image and then decompose this image to create melanin and hemoglobin maps. **b** The cross-polarized image generation module consists of a generator and a discriminator, trained to generate images close to the captured ones. **c** The skin analysis module consists of a skin segmentation model and a skin decomposition model. It identifies the skin regions, divides them into multiple patches, and predicts melanin, hemoglobin, and shading components. During training, these

components are theoretically reconstructed into a skin image, and the model is updated to minimize the difference with the input image. During inference, the melanin and hemoglobin components of each patch are combined to create full-face pigment maps

To primarily focus on verifying the dermatological effectiveness of using generated cross-polarized images in an optical-based trained model, we chose fundamental and lightweight networks. Since each module is independently trained, more advanced architectures can replace the existing structure if developed. Additionally, for clarity, images captured with a visible light source without a polarizer will be referred to as 'standard images' from now on.

Cross-polarized image generation module

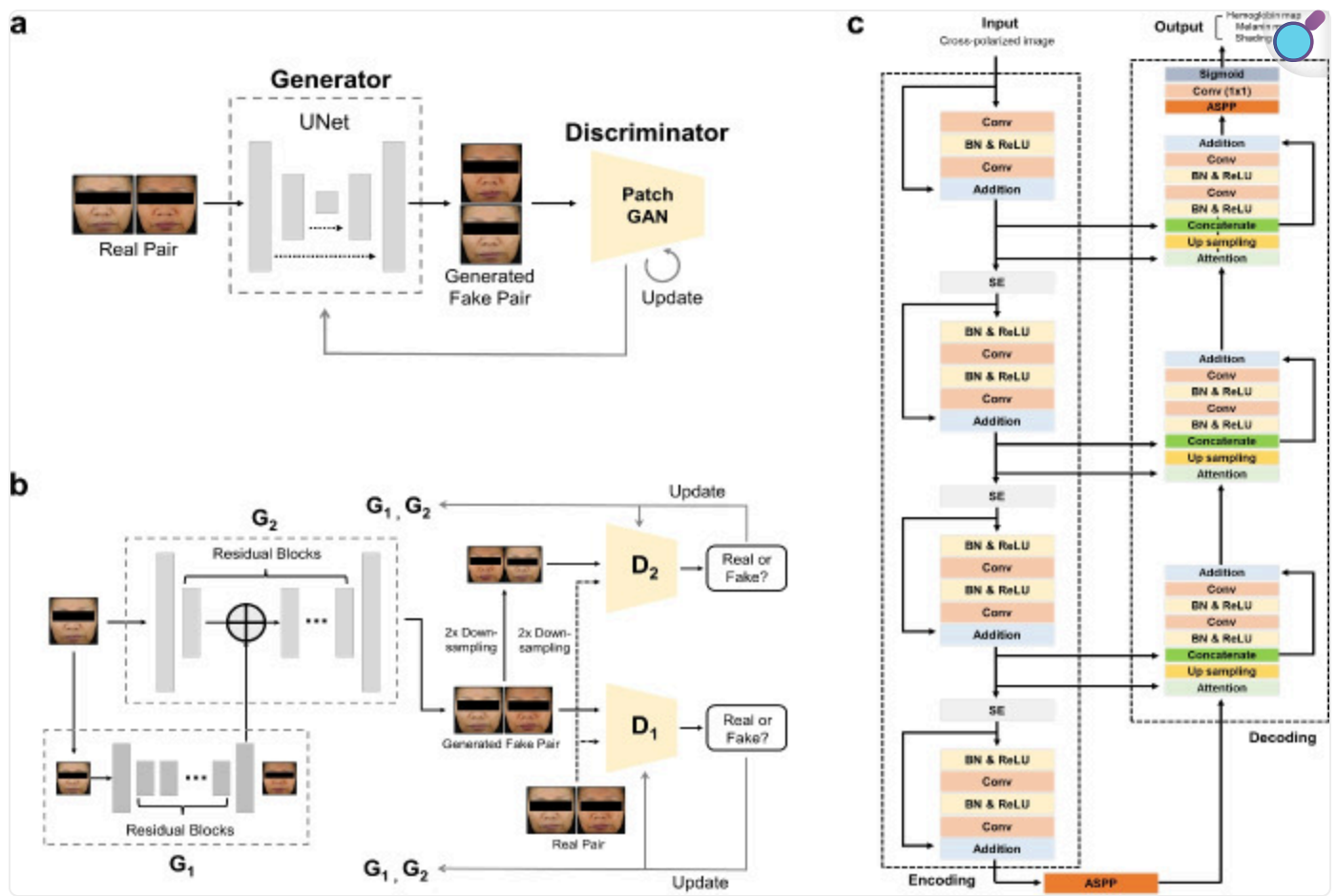
Preparing a high-quality paired dataset of standard images and corresponding cross-polarized images is practically challenging. In contrast, using an unpaired dataset allows the model to autonomously learn the overall features of both domains and alter the image style, which makes data collection easier. However, this approach is highly under-constrained and tends to perform worse than the paired dataset approach, as shown in previous studies [17, 18].

In this study, we conduct a comparative analysis of fundamental models for both paired and unpaired training dataset approaches: pix2pix [19] and CycleGAN [20], respectively, and pix2pixHD [21], an improved version of pix2pix for generating high-resolution images [22].

The architectures of the pix2pix and pix2pixHD models are shown in Fig. 2a and b. The pix2pix model employs a UNet-based generator with skip connections, facilitating effective image-to-image translation by allowing low-level information to bypass the network's deeper layers. Its discriminator is designed as a PatchGAN, classifying various patches of an image as real or fake, and averaging these classifications for a final decision. This approach not only enables a focus on local image details but also helps in maintaining the low-frequency components of the image, thereby enhancing the overall quality [19]. The pix2pixHD model, in contrast, features two sub-networks in its generator: a global generator network (G_1) for low-resolution images and a local enhancer network (G_2) for high-resolution details. The discriminator consists of two identical networks (D_1, D_2) that process images at different scales, simplified from the original model's three-network structure [21]. CycleGAN, another model used in our comparative analysis, utilizes a structure comprising two ResNet-based generators and discriminators using PatchGAN.

However, as CycleGAN was primarily referenced for comparison and not a central focus, its structural diagram is not included in Fig. [2](#).

Fig. 2.



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Network Architectures of pix2pix, pix2pixHD, and ResUNet++ Models. **a** The pix2pix model features a UNet-based generator with skip connections and a convolutional PatchGAN classifier as the discriminator, processing concatenated image pairs. **b** The pix2pixHD model incorporates dual sub-generators: a global generator network (G_1) for low-resolution images and a local enhancer network (G_2) for high-resolution image details. Its discriminator includes two identical networks (D_1 and D_2) for differing image scales, with a downsampling factor of 2. **c** The ResUNet++ architecture comprises a stem block, three squeeze and excitation (SE) blocks, three encoder blocks, Atrous Spatial Pyramid Pooling (ASPP), and three decoder blocks. These include batch normalization and ReLU activation (BN and ReLU), with 3×3 convolution layers

The pix2pix model's objective function integrates a conditional GAN loss with a weighted L1 loss to create realistic target domain images. The optimization involves minimizing the generator's loss while maximizing the discriminator's accuracy, as defined by the following equations [19]:

$$\mathbf{G}^* = \arg \min_G \max_D \mathbf{L}_{cGAN}(G, D) + \lambda \mathbf{L}_1(G) \quad 1$$

$$\begin{aligned} \mathbf{L}_{cGAN}(G, D) = & \mathbf{E}_{x,y} [\log D(x, y)] \\ & + \mathbf{E}_x [\log(1 - D(x, G(x)))] \end{aligned} \quad 2$$

$$\mathbf{L}_1(G) = \mathbf{E}_{x,y} [\|y - G(x)\|] \quad 3$$

G and D are the generator and discriminator, respectively, with λ being the weight of the L1 loss. The GAN loss $\mathbf{L}_{cGAN}(G, D)$ generates realistic outputs but only differentiates between real and fake images, thereby not learning the ground truth directly. To mitigate this issue, the L1 loss $\mathbf{L}_1(G)$ is added to guide the generator, although it tends to produce blurry outputs. Consequently, the optimal ratio for balancing these losses, determined through previous research, is used in this study by setting λ to 100 [19]. The variables x and y represent the paired training images—a standard image and its cross-polarized image.

The pix2pixHD model incorporates a feature matching loss $\mathbf{L}_{FM}(\mathbf{G}, \mathbf{D}_k)$ to leverage multi-scale discriminator features. The model's objective is defined as:

$$\begin{aligned} \mathbf{G}^* = & \min_G ((\max_{D_1, D_2, \dots, D_k} \mathbf{L}_{cGAN}(G, D_k))) \\ & + (\lambda, \sum_{k=1,2,\dots} \mathbf{L}_{FM}(G, D_k),) \end{aligned} \quad 4$$

$$\mathbf{L}_{FM}(G, D_k) = \mathbf{E}_{x,y} \sum_{i=1}^T \frac{1}{N_i} \left[\left\| D_k^{(i)}(x, y) - D_k^{(i)}(x, G(x)) \right\|_1 \right] \quad 5$$

$D_k^{(i)}$ denotes the i -th layer of the k -th discriminator, with T being the total number of layers and N_i the number of elements in each layer.

The pix2pix and pix2pixHD models are trained using the earlier defined objective functions for up to 5000 epochs, with a batch size of 2, employing the Adam optimizer with an initial learning rate of 0.0002. Additionally, CycleGAN is trained on the same dataset for comparative analysis. This model, extending the pix2pix architecture, involves two sets of generator and discriminator models to facilitate image generation using unpaired datasets.

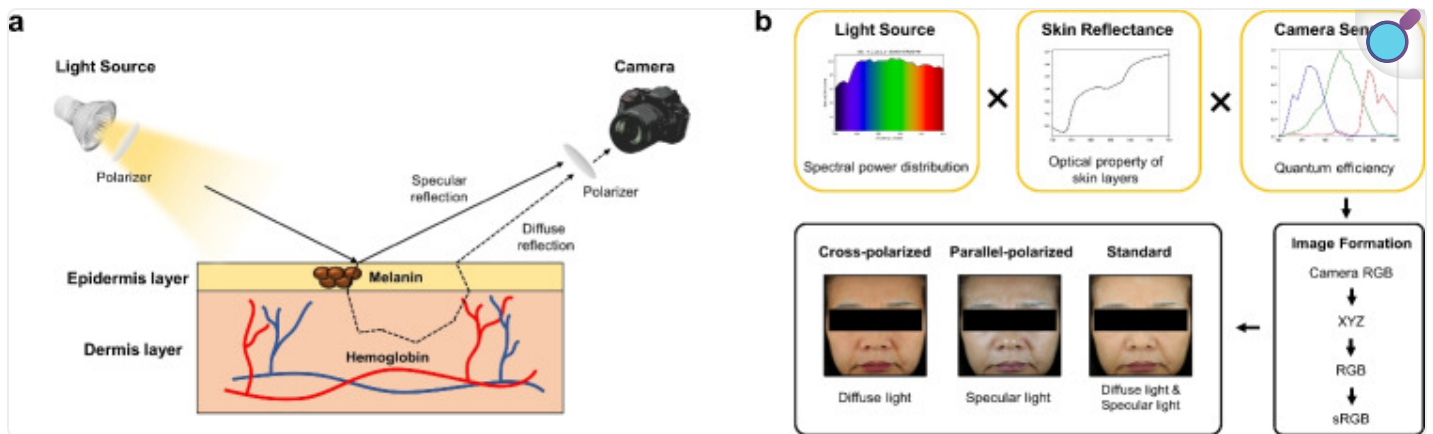
Skin analysis module

To create the ground truth for training the regression network in the skin analysis module, we propose an alternative theoretical method based on the optical properties of pigments. In our prior research [23–26], we developed skin decomposition techniques from computer vision [27] for dermatological applications. While the full-sized image is divided into multiple patches for training, the patch-sized output pigment maps are combined during inference to form full-sized pigment distributions.

Skin image acquisition

Figure 3 illustrates the process of acquiring a skin image. In Fig. 3a, the illuminated light from the light source is partially specularly reflected from the surface due to the refractive index difference between the air and the skin. The remaining light is refracted into the skin layers and is affected by skin pigments, particularly melanin in the epidermis and hemoglobin in the dermis layers. This light is then diffusely reflected back to the air. Both types of reflected light are detected by the camera, resulting in the formation of the standard images. Using a polarizer to distinguish between the two types of reflected light, we can acquire two images: a parallel-polarized image that highlights the skin surface and a cross-polarized image that enhances the skin pigments.

Fig. 3.



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Illustration of the skin image acquisition process. **a** Illuminated light on the skin is either specularly reflected at the surface or diffusely reflected from deeper layers. Both types of reflections contribute to forming a standard image. A polarizer can differentiate these reflections to produce cross-polarized and parallel-polarized images. **b** The formation of skin images is mainly influenced by the light source, skin reflectance, and the camera sensor. Captured image is converted to sRGB image

The image is mainly influenced by the spectral power distribution of the light source, skin reflectance, and the quantum efficiency of the camera sensor, as described in Fig. 3b. Since the created RGB image is in a camera-specific color space, it is converted to XYZ, RGB, and sRGB color spaces for consistent color representation across different devices.

Preprocessing

As the optical characteristics of the skin are employed in the training, the skin segmentation model is used to isolate skin areas while excluding non-skin elements such as the background, eyebrows, eyes, nostrils, and lips. To ensure the broad applicability of this study and other research, the model was trained using standard images and cross-polarized images captured with general cameras and smartphones. The model is based on a lightweight UNet architecture, trained on a dataset of 2117 images and tested on 232 images, achieving a CIoU [28] performance of 0.957.

Given the demand for detailed local information in dermatological applications, high-resolution images are required but need significant memory and computational resources. To manage this, our approach maintains resolution by dividing the image into patches for processing, followed by recombination of the output patches. To mitigate the potential issue of visible borderlines in the combined image, we incorporate overlapping regions between adjacent patches and apply inverse weighting for merging those areas.

For training, patches of 310×310 pixels are utilized, allowing for random cropping as part of data augmentation. The testing phase employs patches of 256×256 pixels, matching the model's input size. A 15-pixel overlap is set between adjacent patches to prevent border discrepancies. Additionally, to prepare the patches for the skin decomposition model, they are converted from the sRGB to RGB colorspace, facilitating a linearized input format.

Skin decomposition model

The skin decomposition model employs the ResUNet++ architecture, advancing beyond UNet and ResUNet by integrating residual blocks, squeeze-and-excitation blocks, Atrous Spatial Pyramid Pooling (ASPP), and an attention block [16] (refer to Fig. 2c). To improve efficiency, we optimized the original ResUNet++ by reducing the number of channels.

This model inputs a cross-polarized image overlaid with a skin segmentation mask, producing melanin, hemoglobin, and shading maps as outputs. Melanin and hemoglobin are the primary skin chromophores, while shading arises from the directional light's geometrical impact. Although a specular component could be included for standard images to capture skin surface features like pores and wrinkles, our study focuses exclusively on diffuse reflection effects in cross-polarized images, thereby simplifying the training process by omitting the specular component [10].

The model's training involves reconstructing output maps into a complete skin image, using a mean squared error (MSE) loss function for optimization and peak signal-to-noise ratio (PSNR) for evaluation. Input images are augmented using the "Albumentations" Python library [29], applying random cropping, flipping, scaling, and rotation to ensure robustness without altering the skin's color.

Forward problem

Visible to near-infrared light penetrates both the epidermis and dermis layers of the skin. The epidermis primarily functions as a melanin absorption layer with minimal scattering due to its thinness, while the dermis combines a hemoglobin absorption layer with a scattering layer composed of collagen and elastin fibers [30] (Fig. 3a). The light-tissue interactions are quantified through absorption and reduced scattering coefficients, represented by the following equations:

$$\mu_a(\lambda) = \sum_i V_i \epsilon_i(\lambda) C_i \quad 6$$

$$\mu'_s(\lambda) = a \left(f_{Ray}, \left(\frac{\lambda}{\lambda_0} \right)^{-4}, +, (1 - f_{Ray}), \left(\frac{\lambda}{\lambda_0} \right)^{-b_{Mie}} \right) \quad 7$$

V_i denotes the volume fraction, ϵ_i the extinction coefficient at wavelength λ , and C_i the concentration of chromophores i , primarily melanin and hemoglobin. The scattering process encompasses both Rayleigh and Mie scattering, with a and b representing the scattering amplitude and power, respectively, and f_{Ray} the fraction of Rayleigh scattering events.

The polarized image acquisition process (illustrated in Fig. 3b) is described by the equation:

$$I_m = \int_{\lambda_{min}}^{\lambda_{max}} I_d(\lambda) C_m d\lambda \quad 8$$

$$I_d(\lambda) = S(\lambda) R(\lambda) M_{shading} \quad 9$$

I_m represents the camera-specific colorspace image for channel $m \in r, g, b$, where $I_d(\lambda)$ is the diffusely reflected light, $C_m(\lambda)$ is the camera sensor's quantum efficiency, and λ ranges from 450 nm to 750 nm. $I_d(\lambda)$ combines the light source's spectral power $S(\lambda)$, skin reflectance $R(\lambda)$, and the shading map $M_{shading}$.

The colorspace of the acquired image is transformed to XYZ and RGB using transformation matrices, and then to sRGB by gamma correction.

Experiments

This study utilized a paired dataset of standard and cross-polarized images obtained with a VISIA-CR imaging system (Canfield Scientific Inc, Fairfield, USA). The dataset, comprising images from

204 subjects, was partitioned into training (175 subjects) and test sets (29 subjects). The dataset was sourced from the Global Medical Research Center, adhering to all relevant national and institutional policies. Participant consent for image use in research and publication was obtained, and the study received approval from the author's institutional review board.

In the cross-polarized image generation module, images with a resolution of 3648×5472 pixels were cropped to remove the background and resized to 1040×1040 pixels for training, allowing for random cropping to 1024×1024 pixels. For testing, images were resized to 1024×1024 pixels after background removal. The module's performance was evaluated using pix2pix, pix2pixHD (trained on paired datasets), and CycleGAN (trained on unpaired datasets) models, with assessments based on PSNR metrics and visual comparison [31].

For skin analysis module, the standard illumination 'D65' spectral power distribution and the 'Canon 5D Mark II' camera quantum efficiency [32] within the 400–720 nm range were referenced. Captured images were compactly cropped and divided into patches (5×4 of 310×310 pixels for training and 6×4 of 256×256 pixels for testing, including overlaps for patch combining). The module's effectiveness was inferred using MSE and PSNR between reconstructed and input images.

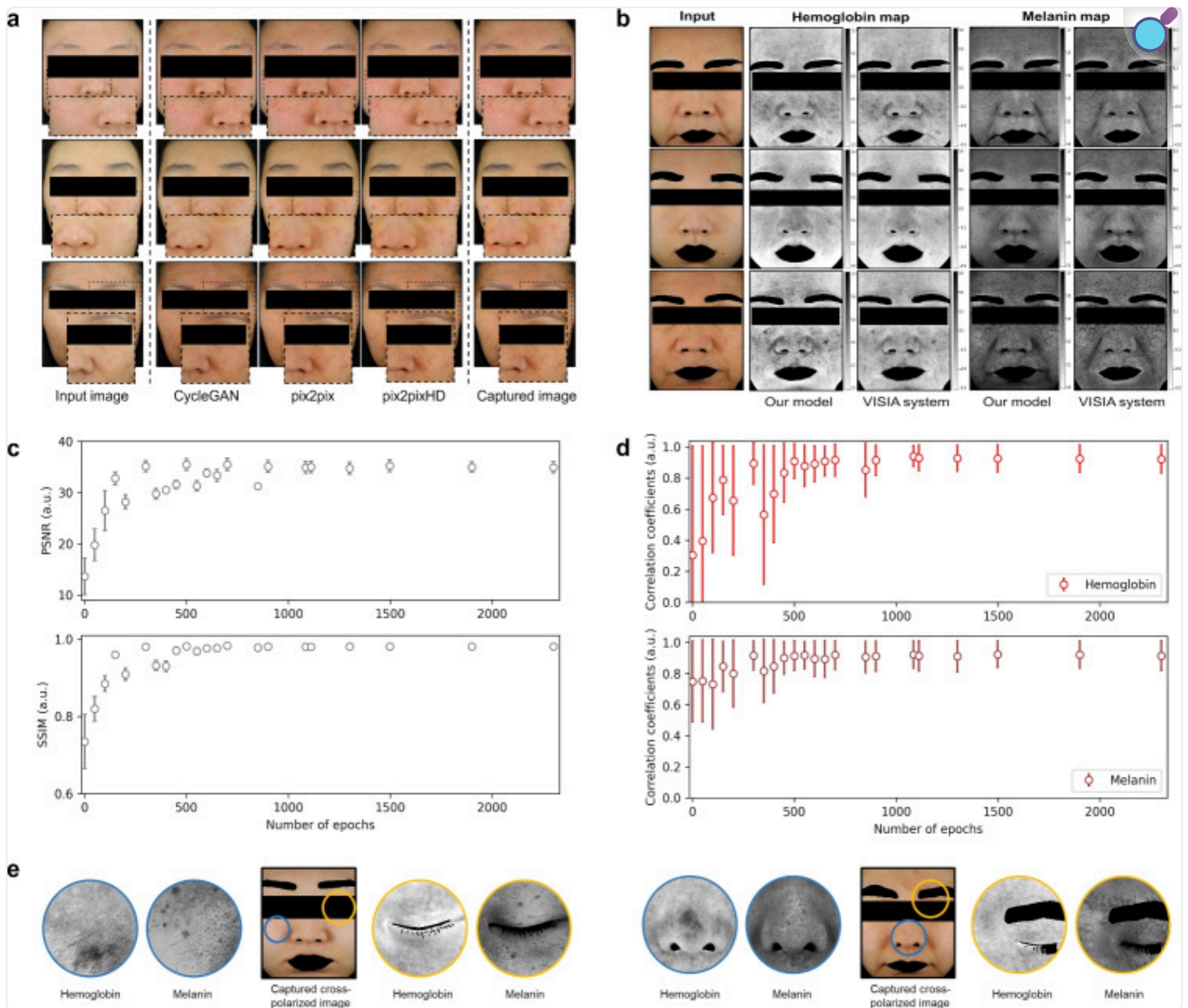
During training, we saved model weights periodically: every 100 epochs for GAN models and every 50 epochs for the skin decomposition model. GAN models can become unstable with prolonged training, making early stopping less effective. Thus, we selected models based on generated images and loss from saved checkpoints, followed by post-processing evaluations to choose the optimal model. For the skin decomposition model, which uses an indirect training approach without ground truth for pigment maps, we initially selected models based on PSNR and then evaluated pigment maps, ultimately choosing the best model based on melanin and hemoglobin correlation coefficients. Additionally, we saved weights whenever the PSNR exceeded 30 dB. To reduce post-processing time, we decreased the sampling frequency after 1000 epochs when changes in PSNR and correlation coefficients were minimal.

The VISIA system results, generated from RBX-Brown and RBX-Red Processing 1.0 module [12], were transformed to grayscale from single-color maps, and compared to our model using Pearson correlation coefficients [33] for melanin and hemoglobin distributions. To minimize potential discrepancies in skin regions between generated and captured cross-polarized images, we cropped the images to include almost the entire face with minimal background and retouched the skin segmentation masks for the test set by manually enlarging regions such as the eyes and mouth to focus on common skin areas.

Results

Figure [4a](#) shows the generated cross-polarized images from the CycleGAN, pix2pix, and pix2pixHD models for the same standard image, with the captured image presented for reference. The PSNR values of the generated images from each model and the captured images for the test set are shown in Table [1](#).

Fig. 4.



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Results of the two modular components: the cross-polarized image generation module and the skin analysis module. **a** Comparison of generated cross-polarized images from three GAN models (CycleGAN, pix2pix, and pix2pixHD) with captured images for reference. **b** Comparison of hemoglobin and melanin maps created by the skin analysis module with those from the VISIA system for the same captured cross-polarized images. **c** Similarity between the reconstructed image and the input image is evaluated using (upper) PSNR and (lower) SSIM for the test dataset (29 subjects). **d** Correlation coefficients of (upper)

hemoglobin and (lower) melanin maps for the same test dataset, between the skin analysis module and the VISIA system. **e** Examples of distinguishing melanin and hemoglobin components from the skin analysis module for qualitative evaluation

Table 1.

The evaluation of the cross-polarized image generation module and the generated image-based integrated approach by comparing to the VISIA system

	CycleGAN	pix2pix	pix2pixHD
Cross-polarized image generation module (PSNR)	30.92 ± 3.07	35.48 ± 1.23	36.53 ± 2.00
Integrated model: Hb (Corr.)	0.904 ± 0.026	0.923 ± 0.027	0.923 ± 0.030
Integrated model: Mel (Corr.)	0.875 ± 0.037	0.879 ± 0.028	0.908 ± 0.027

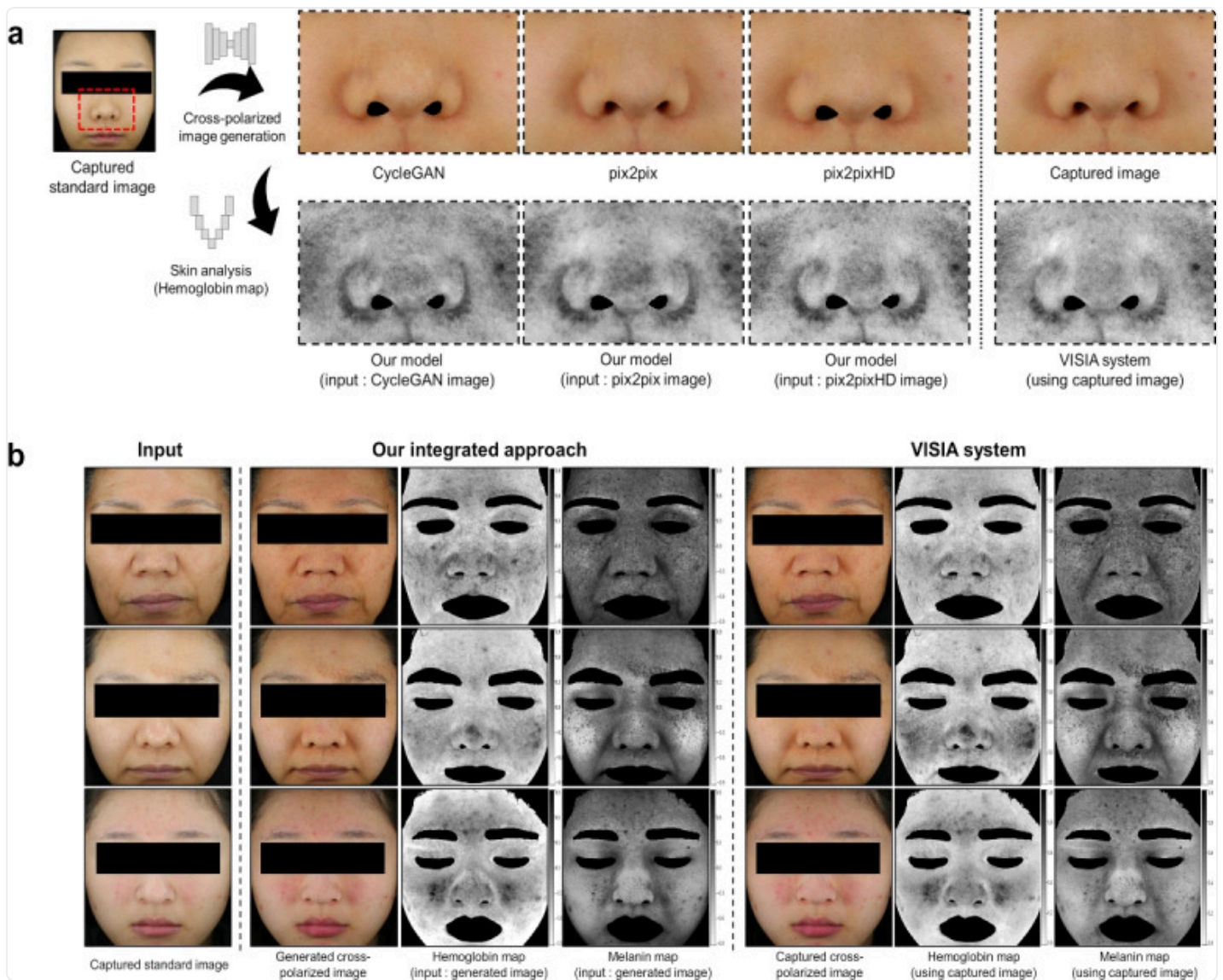
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Figure 4b–e presents the results for the skin analysis module. In Fig. 4b, the melanin and hemoglobin maps from our model and the VISIA system are compared for the same input image, with darker colors representing higher values. The similarity between the input and reconstructed images over training epochs was evaluated using PSNR and Structural Similarity Index Measure (SSIM), as shown in Fig. 4c. The maximum values for PSNR and SSIM are 35.514 dB (at 500 epochs) and 0.983 (at 700 epochs), respectively. The correlation coefficients between the pigment maps from the skin analysis module and the VISIA system over training epochs are shown in Fig. 4d, with the highest values for hemoglobin and melanin being 0.942 and 0.922, respectively. Figure 4e shows examples of distinguishing melanin and hemoglobin components for qualitative evaluation.

The results of the integrated approach are shown in Fig. 5 and Table 1. For all cases, our system generated cross-polarized images, which were then input into our skin analysis module (hemoglobin: 0.942, melanin: 0.922) to create pigment maps, while the VISIA system analyzed

captured images using their commercial tool. The generated images from the three GAN models (CycleGAN, pix2pix, and pix2pixHD) and their corresponding hemoglobin maps created by our module are compared in Fig. 5a. For the test set, the correlation coefficients between the pigment maps created by the integrated approach and those from the VISIA system are shown in Table 1. Through qualitative and quantitative comparisons, we determined that the pix2pixHD model performed the best. Figure 5b compares the pix2pixHD-based generated images and the created pigment maps for a full face with those captured and analyzed by the VISIA system.

Fig. 5.



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Results of the integrated approach. **a** Magnified cross-polarized images generated by three GANs (CycleGAN, pix2pix, and pix2pixHD) and the corresponding hemoglobin maps created by the skin analysis module for these images. For reference, the captured image and its analyzed hemoglobin map by the VISIA system are included. **b** Comparison of our integrated approach (pix2pixHD-based) and the VISIA system (captured image-based) for the same full-face standard image

Discussion

The separate modules were evaluated first, and then their integrated approach was assessed. For the cross-polarized image generation, the redness and blemish areas, which are related to hemoglobin and melanin, are enhanced in the images generated by the pix2pix and pix2pixHD models, whereas those from the CycleGAN model showed relatively less change (Fig. 4a). Similarly, the PSNR of the pix2pix and pix2pixHD models are above 35 dB, indicating that the generated images are highly similar to the captured images and difficult to distinguish with the human eye [34], while that of the CycleGAN model is around 30 dB (Table 1). The unpaired dataset of 175 subjects might not be sufficient for the CycleGAN model to find latent features for conversion between the two domains. However, previous studies have shown that the performance of the pix2pix model, which learns from paired-pixel level information, is better than that of the CycleGAN, even with the same training dataset [20]. While the pix2pix and pix2pixHD models generate qualitatively similar images, the pix2pixHD model achieves a PSNR that is 1.05 dB higher (Table 1), likely due to its enhancements for generating higher-resolution images. To achieve a higher PSNR than this, it is necessary to overcome practical challenges in image acquisition. The standard image and cross-polarized image were taken sequentially, making it difficult to avoid motion artifacts due to pixel-level movement. This issue can be resolved by either performing precise post-processing or using a beam splitter to capture both images simultaneously when acquiring images.

For the skin analysis module, it shows a similar qualitative tendency in melanin and hemoglobin distributions to the VISIA system (Fig. 4b). Even in regions where melanin and hemoglobin overlap, the module distinguishes these components (Fig. 4e). Vessels over the eyes (left image) and redness regions on the nose (right image) are classified as hemoglobin, while moles and black spots in the same regions are classified as melanin. The PSNR and SSIM of the reconstructed images over training epochs (Fig. 4c) and the correlation coefficients of the pigment maps (Fig. 4d) remain consistent without significant change after approximately 500 epochs. The best model for creating pigment maps (at 1085 epochs) shows that the correlation coefficient of hemoglobin (0.942) is higher than that of melanin (0.922). This result is interpreted as the difficulty in distinguishing between melanin and shading maps due to their shared dark color characteristics when trained without ground truth.

In addition, the reasons for using PSNR and SSIM to evaluate the similarity between the reconstructed and input images are as follows. While PSNR provides a straightforward numerical comparison of the overall fidelity between the generated and reference images by measuring the difference in pixel values, it does not effectively reflect human visual perception [31]. In contrast,

SSIM evaluates perceptual quality by considering luminance, contrast, and structural similarities, resulting in a measure that is close to visual perception. By incorporating both PSNR and SSIM, we ensure a more comprehensive assessment of the reconstructed image quality. In case of GAN models, while Frechet Inception Distance (FID) and Inception Score (IS) are commonly used evaluation metrics, obtaining unbiased and reliable results from these metrics requires a large number of datasets [35]. Therefore, they are difficult to apply in this study.

For the integrated approach, we compared three GAN models in Fig. 5a. Although the overall tendency of the hemoglobin maps created by our model is similar to that of the VISIA system, they appear slightly blurred. There could be two possible reasons for this: GANs tend to produce intermediate values to stably minimize the loss function, which can result in the loss of detailed information and introduce blurring [19]. Additionally, since the VISIA system analyzes high-resolution images captured by a DSLR camera, it preserves more detailed characteristics compared to the images generated by GANs. Furthermore, when using GAN (pix2pixHD) generated images as input for the skin analysis module, the created hemoglobin map decreased by 2.1% (0.942–0.923) and the melanin map decreased by 2.6% (0.922–0.897) compared to using captured images as input (Table 1). Considering that the comparison is made against a professional analysis tool using their image acquisition system, even the decreased values are still considered positive. When comparing full images (Fig. 5b), the hemoglobin maps for the first and third subjects are partially over-emphasized compared to the VISIA system. This could be due to excessive redness in the generated cross-polarized image or differences in scale during the plotting process of the pigment maps.

Several methods can be considered for further improving the proposed approach and transitioning it into clinical practice. First, since the approach has a modular structure, each component can be replaced with more enhanced models. Using an improved GAN model capable of generating more detailed information could significantly improve the analysis results of the integrated approach. Another possible improvement is to include shading ground truth in training, which will enhance the ability to distinguish between melanin and shading. Additionally, incorporating additional deep learning models that adapt to various image acquisition conditions could greatly increase the versatility of the proposed approach across different image acquisition systems.

For clinical application, extensive clinical trials are essential to validate the effectiveness of the proposed method. These trials should involve diverse skin pigment distributions and concentrations-such as skin tone, race, and pigment-related diseases-to ensure the model's general applicability. Furthermore, to extend our study from creating pigment maps to classifying diseases,

enhancing the explainability of the AI models will build trust with clinicians and ensure the method's transparency [36].

In this study, we proposed an integrated approach for generating cross-polarized images and creating skin melanin and hemoglobin maps. There have been no similar prior studies in this area. It can generate images with enhanced melanin and hemoglobin areas without using additional optical instruments to acquire cross-polarized images. Moreover, it can produce results similar to those of professional tools without needing ground truths. In the future, we hope that our approach can be used as a principle to replace existing professional dermatological equipment by ensuring the model's general applicability through large datasets composed of subjects with various conditions.

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Author contributions

Geunho Jung was involved in writing the manuscript and implemented a skin analysis module. Jongha Lee was involved in conceptualization of the main idea, project management, and preparing datasets from external organizations. Semin Kim made technical contributions in implementing deep learning models and interpreting the results. All authors contributed to data analysis and reviewed the manuscript.

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Code and data availability

Due to proprietary technology and privacy constraints, the code and data associated with this article are not publicly available. However, we are open to discussion for research purposes related to the code's development, and data requests can be made through official collaborations with our company. Such requests should contact the corresponding author, Semin Kim.

Declarations

Conflict of interest

The authors declare that they have no relevant financial interests in this article and no potential Conflict of interest to disclose.

Ethical approval

The research data was provided by GMRC (Global Medical Research Center, Republic of Korea) and the IRB approval number is 'GIRB-21426-PD'.

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References

1. Atef A, El-Rashidy MA, Azeem AA, Kabel AM. The role of stem cell factor in hyperpigmented skin lesions. *Asian Pacific J Cancer Prev: APJCP*. 2019;20(12):3723. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
2. Troyanova P, Borisova E, Avramov L. Fluorescence and reflectance properties of hemoglobin-pigmented skin disorders. In: *International Conference on Lasers, Applications, and Technologies 2007: Laser Technologies for Medicine*. vol. 6734. SPIE; 2007. pp. 142–149.

3. Gong H, Desvignes M. Hemoglobin and melanin quantification on skin images. In: Image Analysis and Recognition: 9th International Conference, ICIAR 2012, Aveiro, Portugal, June 25-27, 2012. Proceedings, Part II 9. Springer; 2012. pp. 198–205.
4. Goldsberry A, Hanke CW, Hanke KE. VISIA system: a possible tool in the cosmetic practice. *J Drugs Dermatol: JDD*. 2014;13(11):1312–4. [[PubMed](#)] [[Google Scholar](#)]
5. Linming F, Wei H, Anqi L, Yuanyu C, Heng X, Sushmita P, et al. Comparison of two skin imaging analysis instruments: the VISIA from Canfield vs. the ANTERA 3D CS from Miravex. *Skin Res Technol*. 2018;24(1):3–8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
6. Dobrev H. Fluorescence diagnostic imaging in patients with acne. *Photodermatol, Photoimmunol Photomed*. 2010;26(6):285–9. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
7. Zonios G, Bykowski J, Kollias N. Skin melanin, hemoglobin, and light scattering properties can be quantitatively assessed in vivo using diffuse reflectance spectroscopy. *J Invest Dermatol*. 2001;117(6):1452–7. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
8. Jung B, Choi B, Durkin AJ, Kelly KM, Nelson JS. Characterization of port wine stain skin erythema and melanin content using cross-polarized diffuse reflectance imaging. *Lasers Surg Med*. 2004;34(2):174–81. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
9. Kojima K, Shido K, Tamiya G, Yamasaki K, Kinoshita K, Aiba S. Facial UV photo imaging for skin pigmentation assessment using conditional generative adversarial networks. *Sci Rep*. 2021;11(1):1213. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
10. Tsumura N, Ojima N, Sato K, Shiraishi M, Shimizu H, Nabeshima H, et al. Image-based skin color and texture analysis/synthesis by extracting hemoglobin and melanin information in the skin. In: *ACM SIGGRAPH 2003 Papers*. Association for Computing Machinery; 2003. pp. 770–779.
11. Jakovels D, Spigulis J, Sakniete I. Multi-spectral mapping of in vivo skin hemoglobin and melanin. In: *Biophotonics: Photonic Solutions for Better Health Care II*. vol. 7715. SPIE; 2010. pp. 575–580.
12. Demirli R, Otto P, Viswanathan R, Patwardhan S, Larkey J. RBX® technology overview. *Canfield Syst White Pap*. 2007;1:1–5. [[Google Scholar](#)]
13. Tsumura N, Haneishi H, Miyake Y. Independent-component analysis of skin color image. *JOSA A*. 1999;16(9):2169–76. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

14. Liu Y, Jain A, Eng C, Way DH, Lee K, Bui P, et al. A deep learning system for differential diagnosis of skin diseases. *Nat Med*. 2020;26(6):900–8. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
15. Kaymak R, Kaymak C, Ucar A. Skin lesion segmentation using fully convolutional networks: a comparative experimental study. *Expert Syst Appl*. 2020;161: 113742. [[Google Scholar](#)]
16. Jha D, Smedsrud PH, Riegler MA, Johansen D, De Lange T, Halvorsen P. Resunet++: An advanced architecture for medical image segmentation. In: *IEEE International Symposium on Multimedia (ISM)*. IEEE; 2019. pp. 225–2255.
17. Singh NK, Raza K. Medical image generation using generative adversarial networks: A review. *Health informatics: A computational perspective in healthcare*; 2021. pp. 77–96.
18. Tripathy S, Kannala J, Rahtu E. Learning image-to-image translation using paired and unpaired training samples. In: *Computer Vision–ACCV 2018: 14th Asian Conference on Computer Vision, Perth, Australia, December 2–6, 2018, Revised Selected Papers, Part II* 14. Springer; 2019. pp. 51–66.
19. Isola P, Zhu JY, Zhou T, Efros AA. Image-to-image translation with conditional adversarial networks. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. IEEE; 2017. pp. 1125–1134.
20. Zhu JY, Park T, Isola P, Efros AA. Unpaired image-to-image translation using cycle-consistent adversarial networks. In: *Proceedings of the IEEE international conference on computer vision*. IEEE; 2017. pp. 2223–2232.
21. Wang TC, Liu MY, Zhu JY, Tao A, Kautz J, Catanzaro B. High-resolution image synthesis and semantic manipulation with conditional gans. In: *Proceedings of the IEEE conference on computer vision and pattern recognition*. IEEE; 2018. pp. 8798–8807.
22. Chen Q, Koltun V. Photographic image synthesis with cascaded refinement networks. In: *Proceedings of the IEEE international conference on computer vision*. IEEE; 2017. pp. 1511–1520.
23. Jung G, Lee J, Kim S. Spectrum-based deep learning framework for dermatological pigment analysis and simulation. *Comput Biol Med*. 2024;178: 108741. 10.1016/j.compbiomed.2024.108741. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]

24. Jung G, Kim S, Lee J, Yoo S. Deep learning-based optical approach for skin analysis of melanin and hemoglobin distribution. *J Biomed Optics*. 2023;28(3): 035001. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
25. Jung G, Kim S, Lee J, Yoo S. Deep learning-based pigment analysis model trained with optical approach and ground truth assistance. *J Biophotonics*. 2023;16(12): e202300231. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
26. Jung G, Kim S, Lee J, Yoo S. Generation of skin tone and pigmented region-modified images using a pigment discrimination model trained with an optical approach. *Skin Res Technol*. 2023;29(10): e13486. [[DOI](#)] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#)]
27. Alotaibi S, Smith W. Biofacenet: Deep biophysical face image interpretation. arXiv preprint [arXiv:1908.10578](#). 2019.
28. Xu C, Wang J, Yang W, Yu L. Dot distance for tiny object detection in aerial images. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. IEEE; 2021. pp. 1192–1201.
29. Buslaev A, Iglovikov VI, Khvedchenya E, Parinov A, Druzhinin M, Kalinin AA. Albumentations: fast and flexible image augmentations. *Information*. 2020;11(2):125. [[Google Scholar](#)]
30. Dolotov L, Sinichkin YP, Tuchin V, Utz S, Altshuler G, Yaroslavsky I. Design and evaluation of a novel portable erythema-melanin-meter. *Lasers Surg Med*. 2004;34(2):127–35. [[DOI](#)] [[PubMed](#)] [[Google Scholar](#)]
31. Setiadi DRIM. PSNR vs. SSIM: imperceptibility quality assessment for image steganography. *Multimedia Tools Appl*. 2021;80(6):8423–44. [[Google Scholar](#)]
32. Jiang J, Liu D, Gu J, Süsstrunk S. What is the space of spectral sensitivity functions for digital color cameras? In: *IEEE Workshop on Applications of Computer Vision (WACV)*. IEEE; 2013, pp. 168–79.
33. Puth MT, Neuhäuser M, Ruxton GD. Effective use of Pearson's product-moment correlation coefficient. *Anim Behav*. 2014;93:183–9. [[Google Scholar](#)]
34. Lee CC, Wu HC, Tsai CS, Chu YP. Adaptive lossless steganographic scheme with centralized difference expansion. *Pattern Recogn*. 2008;41(6):2097–106. [[Google Scholar](#)]

35. Borji A. Pros and cons of GAN evaluation measures: new developments. *Comput Vis Image Underst.* 2022;215: 103329. [[Google Scholar](#) 

36. Chanda T, Hauser K, Hobelsberger S, Bucher TC, Garcia CN, Wies C, et al. Dermatologist-like explainable AI enhances trust and confidence in diagnosing melanoma. *Nat Commun.* 2024;15(1):524. [[DOI](#) ] [[PMC free article](#)] [[PubMed](#)] [[Google Scholar](#) 

Associated Data

This section collects any data citations, data availability statements, or supplementary materials included in this article.

Data Availability Statement

Due to proprietary technology and privacy constraints, the code and data associated with this article are not publicly available. However, we are open to discussion for research purposes related to the code's development, and data requests can be made through official collaborations with our company. Such requests should contact the corresponding author, Semin Kim.

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